Sukanya Krishna

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Education

Harvard University **2024-**
Doctor of Philosophy (PhD) in Computer Science; Secondary Field in Mind, Brain, and Behavior *Cambridge, MA*

University of California, San Diego (UCSD) **2024**
Bachelor of Science (BS) in Bioengineering; Double Minor in Data Science and Cognitive Science *San Diego, CA*

- GPA: 3.995/4.0 (summa cum laude, Valedictorian of graduating class)

Select Graduate Research Projects

AI for Social Impact **09/2024 – Present**
Harvard SEAS; Harvard Medical School; Boston Children's Hospital *Mentor(s): Prof. Maimuna Majumder; Prof. Finale Doshi-Velez*

- Working on various projects developing computational and reinforcement learning–based models of human behavior to study decision-making, uncertainty, and human-AI complementarity including applications in high-stakes health settings.
- Designing epistemically aware AI systems for public health, including a substance overdose excess mortality dashboard for policymakers and uncertainty-calibrated behavioral intervention models.

Epistemic Integrity and Trust in AI **06/2024 - Present**
Harvard SEAS; Mila - Quebec AI Institute *Mentor(s): Prof. Reihaneh Rabbany; Prof. Finale Doshi-Velez*

- Developing a conversational negotiation sparring partner that analyzes proposed strategies and supporting documents, identifies loopholes, inconsistencies, and vulnerabilities, and suggests stronger strategic alternatives (work as part of Cosmos Grant 2026)
- Investigating epistemic integrity in large language models (LLMs) to understand how humans respond to confidence signals in communication. Using our Veracity application as the user-facing application for the ongoing study.
- Worked on Veracity, a conversational assistant that will help clarify the veracity or reliability of factual claims

AI Companions in Medicine **09/2024 - Present**
Amsterdam UMC *Advisor: Dr. Sanjat Kanjilal*

- Evaluating local LLMs for extraction and summarization of Infectious Disease Syndromes from Multilingual Electronic Health Records. Running a bilingual, two-center study between BWH and Amsterdam UMC
- Built an AI-driven medical course tutor that uses LLMs to generate case studies and engage students in Socratic dialogue. Running pilot in HST40 course (Mechanisms of Microbial Pathogenesis) in Fall 2026

Assessing Diversifiable Risk in AI Infrastructure Investments **04/2026 - 8/2026**
Harvard T.H. Chan School *Mentor(s): Prof. Francesca Dominici*

- We introduce infrastructure observability as a missing layer in technical AI governance and propose a quantitative framework to assess diversifiable risks in AI infrastructure investments
- We identify and quantify site-level risks given the following: financial opacity, geographic vulnerability, resource stress, community impact, and market mismatches. We combine these risks into portfolio-level exposure metrics that capture the cumulative risk profile of a collection of infrastructure assets held by an investor

Human-AI Complementarity for Amplified Oversight **09/2025 - 06/2026**
Supervised Program for Alignment Research (SPAR) *Mentor: Dr. Rishub Jain (Google DeepMind)*

- Developed hybridization and AI-assistance frameworks to improve human-AI team accuracy on reasoning and factuality tasks, using confidence calibration and sub-task delegation.
- Built and evaluated a multi-domain oversight dataset (e.g., QuALITY, BIG-Bench, GPQA-Diamond) to test complementarity between human and model judgments.
- Implemented confidence-based decision mechanisms and interactive assistance tools to study how human-AI collaboration can enhance calibration and oversight reliability.

Coordination Vulnerabilities in LLM Multi-Agent Systems **06/2025 - 09/2025**
Pivotal Research @ London Institute for Safe AI (LISA) *Mentor: Dr. Tobin South (Stanford; Anthropic)*

- Investigated vulnerabilities in large language model (LLM) multi-agent coordination, focusing on how contextual framing interacts with explicit strategic advice across social dilemmas (Prisoner's Dilemma, Battle of the Sexes, Tragedy of the Commons).

Select Industry Experiences

IBM Research **05/2026 - Present**
Research Collaborator on the IT Automation Team *Mentor(s): Dr. Raju Pavuluri, Dr. Yu Deng, Dr. Rahul Krishna, Dr. Saurabh Jha*

- Developing benchmark (INFERBench) for evaluating whether AI agents can reason about and identify vulnerability chains that span realistic enterprise systems rather than a single vulnerable program
- Unlike prior benchmarks that focus on application-level exploits or isolated infrastructure misconfigurations, INFERBench models interactions across application code, secrets and identity artifacts, infrastructure-as-code, cloud/IAM configuration, network policies, protocol behavior, and emerging agentic/API trust failures

Lexi AI, Inc. **03/2025 - 3/2026**
Part time Machine Learning Engineer (NLP + Speech)

- Work part-time on a flexible, as-needed basis at Harvard startup focused on delivering real-time AI medical interpretation to the 30 million patients navigating language barriers in healthcare.

- Helped with efforts to fine-tune and evaluate NLP and speech models (e.g., speech-to-text, text-to-text translation, text-to-speech)
- Curated and preprocessed multilingual datasets, particularly for medical and low-resource domains
- Set up benchmarking pipeline to benchmark model performance across languages and modalities (e.g., accuracy, latency, and clinical relevance)
- Collaborating with our technical and clinical teams to improve model integration, safety, and equity

Amazon (One Medical) **5/2023 - 8/2023**

Product Analytics Intern

- Developed an efficient pre-aggregated data model on Snowflake using SQL, aggregating patient data to enhance performance and analytical capabilities.
- Designed and published Tableau dashboards, visually representing 8 crucial success metrics sourced from Snowflake and Mixpanel.

Medtronic **6/2022 - 8/2022**

Data Engineering Intern

- Reduced compute time on Digital Twin for real-time intervention (50k patients on InPen therapy dataset)
- Achieved 7.5x reduction in compute time with stable fitting (1.4% deviation in MARD), and stable parameter estimation (less than 4% parameter variation) for 10-minute step size
- Estimated 5x cost reduction in cloud resources (AWS) at scale

Conference/Journal/Workshop Papers

- Sallami, D., **Krishna, S.**, ... & Rabbany, R. (2026). "VeracityExt: In-Browser Fact-Checking." *Submitted to* The 40th Annual AAAI Conference on Artificial Intelligence (AAAI '27) Demonstrations Paper Track
- Shi, S., Christ, M., ..., **Krishna, S.**, ... & Jain, R. (2026). "Towards Human-AI Complementarity for AI Safety." *Submitted to* The 25th International Conference on Learning Representations (ICLR '27)
- **Krishna, S.***, Sallami, D.*, ... & Doshi-Velez, F. (2026). "Rethinking Automated Fact-Checking Evaluation: Disentangling Input Ambiguity from System Failure." *Submitted to* The 40th Annual AAAI Conference on Artificial Intelligence (AAAI '27) Special Track on Artificial Intelligence for Social Impact
- Sidkar, S.*, **Krishna, S.***, ... & Majumder, M. (2026). "How Pandemic Publishing Built Higher Walls, Not Bridges." *Submitted to* the 33nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '27)
- **Krishna, S.***, Sallami, D.*, ... & Doshi-Velez, F. (2026). "Fact-Checking Evaluation Must be Ambiguity-Aware." *Submitted to* The 40th Annual Conference on Neural Information Processing Systems (NeurIPS '26) Position Paper Track
- Karagozian, B.*, **Krishna, S.***, ... & Dominici, F. (2026). "Towards Infrastructure-Aware AI Governance: Assessing Diversifiable Risk in AI Infrastructure Investments." *Submitted to* The 40th Annual Conference on Neural Information Processing Systems (NeurIPS '26) Position Paper Track
- Zheng, Y., Dahmani, A., ..., **Krishna, S.**, ... & Jain, R. (2026). "Confidence-Aware Human-AI Complementarity." 2026 NSF AI-SDM Human-AI Complementarity Workshop: Dynamic Alignment
- **Krishna, S.**, Perrault, A., Charpignon, M., Majumder, M. (2026). "Spatiotemporal Modeling of the Substance Overdose Epidemic in the United States Using Time-Series Clustering." Workshop Paper. The 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '26). epiDAMIK Workshop
- **Krishna, S.**, South, T. (2026). "Beyond Vibe Decision Theory: Asymmetric Manipulation Vulnerabilities in LLM Multi-Agent Coordination" Research Paper. In Proceedings of The 25th International Conference on Autonomous Agents and Multiagent Systems (AAMAS '26). May 25-29 2026. Paphos, Cyprus
- **Krishna, S.**, Charpignon, M., Majumder, M. (2025). "Statistical vs. Deep Learning Models for Estimating Substance Overdose Excess Mortality in the US." arXiv preprint arXiv:2512.21456
- **Krishna, S.**, South, T. (2025). "Asymmetric Manipulation Vulnerabilities in LLM Multi-Agent Coordination" Workshop Paper. The 39th Annual Conference on Neural Information Processing Systems (NeurIPS '25). Workshop on Algorithmic Collective Action (ACA). Dec. 2-7, 2025. San Diego, CA
- Curtis, T., Touzel, M., ..., **Krishna, S.**, ... & Pelrine, K. (2025). "Veracity: An Open-Source AI Fact-Checking System." Demo Paper (Demo Video). The International Joint Conference on Artificial Intelligence (IJCAI '25) AI And Social Good (Special Track). August 16-22, 2025. Montreal, Canada
- **Krishna, S.**, Martinez, J., Kim, S., ... & Morimoto, T. (2025). "Towards Soft Steerable Vine Robots Using Actuating Polymers." Research Paper. In Proceedings of The Institute of Electrical and Electronics Engineers International Conference on Soft Robotics (IEEE RoboSoft '25). April 23-26, 2025. Lausanne, Switzerland
- Ghafouri, B., Mohammadzadeh, S., ..., **Krishna, S.**, ... & Pelrine, K. (2024). "Epistemic Integrity in Large Language Models." Workshop Paper. The 38th Annual Conference on Neural Information Processing Systems (NeurIPS '24). Safe Generative AI Workshop. December 10-15, 2024. Vancouver, Canada
- Tsan, S., Kansal, R., ..., **Krishna, S.**, ... & Pierini, M. (2021). "Particle Graph Autoencoders and Differentiable, Learned Energy Mover's Distance." Workshop Paper. The Annual Conference on Neural Information Processing Systems (NeurIPS '21). Machine Learning for the Physical Sciences Workshop. December 6-14. Virtual, USA

Service Activities, Leadership & Teaching

- **Leadership & Engagement:**
  - Student Leader in AI at the Berkman Klein Center (SLAI)
  - Policy Fellow and Member of AI Safety Student Team (AISST)
  - Student Leader for the Harvard Interfaith Dialogue Program
  - SEAS InTouch Peer Mentor, fostering graduate community and well-being
- **Conference Organization:** Organizing Committee Member for the First Workshop on Agent Evaluation Science (AgentEvalSci) at NAACL 2027.
- **Peer Reviewing:** Have been member of program committee for The 2026 ACM Web Conference The Future of Information Integrity Research Workshop (FIIR '26), The IEEE International Conference on Data Mining (ICDM) Future of Information Integrity Reserach Workshop (FIIR '26), and The 2025/2026 IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI'25, BHI'26)
- **Harvard Bok Graduate Fellow** (2026-2027) - support innovative course development within the department, partnering with faculty and Bok Center staff to design a variety of tools, course materials, and assignments for undergraduate courses and departments/programs
- **Teaching Experience:** Graduate Teaching Assistant for *Advanced Topics in Data Science 1* (Fall 2026), *Advanced Topics in Data Science 2* (Spring 2026), and *Introduction to Scientific Computing* (Fall 2025) at Harvard; previously Instructional Assistant for *Computational Methods in Engineering*, *Machine Learning II*, and *Linear Circuits* at UCSD; former Ignite Fellow with Teach for America and content creator for UCSD's *Big BENG* educational initiative.

Select Fellowships, Awards, and Honors

- **Cosmos Institute AI for Truth-Seeking Research Grant (2026)** - won a $6,000 independent grant to support research related to developing technology that supports the open contestation of ideas and human judgment in knowledge production
- **GEM Employer Fellow (2026)** - fellowship from The National GEM Consortium towards 2026-27 academic year ($20,000). Sponsored by IBM Research
- received support from the Porthcawl Innovation Fund in the Division of Engineering and Applied Sciences to fund year 2025-26 of PhD (2025)
- **Harvard School of Engineering and Applied Sciences (SEAS) Prize Fellowship (2024)** - Awarded to select incoming students across Harvard Graduate School of Arts and Sciences to cover full tuition and educational fees (≈$60,000/yr)
- Outstanding Academic Achievement Award in Bioengineering at UCSD (2024)
- Rotary Club of Murrieta Academic Undergraduate Scholarship (2020) - for involvement in community service activities during high school ($1000/yr)
- Institute for Research and Innovation in Software for High Energy Physics (IRIS-HEP) Undergraduate Fellowship (2021) - summer research fellowship
- UCSD Thurgood Marshall College Provost Honors (2020-2024)